GGG-C-555

DECEMBER 8, 1955

SUPERSEDING
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1 October 1954

FEDERAL SPECIFICATION

COMPRESSOR, PISTON RING, BAND-TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — The piston ring compressors covered by this specification are used to firmly compress the piston rings into the grooves of the piston to facilitate insertion of the piston and rings into the cylinder.

1.1.1 *Federal specification coverage.* — This specification does not include all types and classes of piston ring compressors, but is intended to cover only those types and classes generally used by the Federal Government.

1.2 Classification.

1.2.1 *Sizes.* — Piston ring compressors shall be of the following sizes, as specified (see 6.1):
Size 1.—2⅛- to 5-inch capacity.
Size 2.—3½- to 7-inch capacity.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specification and standard of the issues in effect on date of invitation for bids, form a part of this specification:

Military Specification:
MIL-P-15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:
MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustration. — The illustration shown herein is for convenience of identification and does not preclude purchase of piston ring compressors otherwise conforming to this specification.

3.2 Material. — The material used in construction of the piston ring compressors shall be as hereinafter specified. The material shall be free from any defects and imperfections that may affect their serviceability. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose intended.

3.3 Design. — The piston ring compressors shall consist essentially of a compression band, two tension bands, spool, ratchet, pawl and a

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removable key or wrench. The piston ring compressors shall be designed to accomplish the following:

(a) To compress the heaviest piston rings serviced by means of twin band tension providing equalized pressure over the width of the compression band.

(b) To provide the compression force by means of a spool, actuated by either a removable key or wrench.

(c) To lock in position at any desired point within the operating range by means of a ratchet and pawl.

(d) To provide takeup tension of approximately 0.010 inch per tooth on ratchet.

(e) To provide a friction brake for controlling opening feature.

(f) To provide suitable means on the lower edge of the compression band to prevent the compression band from entering cylinder when piston is inserted.

3.3.1 *Compression band.* — The compression band shall be of good quality spring steel, properly tempered so as to fully open the completely assembled compressor to the largest required diameter when not restrained. The width of the compression band shall be not less than 3¼ inches.

3.3.2 *Tension bands.*—The two tension bands shall be of good quality spring steel, properly tempered, and of ample strength to compress the heaviest rings to be serviced. Means shall be provided for proper assembly of the tension bands to the spool.

3.3.3 *Spool.* — The spool shall be of good quality steel, designed to properly accept and wind the tension bands without slippage. Means shall be provided in the spool to properly accept a removable key or wrench which actuates the spool.

3.3.4 *Ratchet.*—The ratchet shall be of good quality steel, properly heat-treated to resist wear or breakage under normal service conditions. The ratchet shall be designed for proper assembly to the spool and shall have the teeth so spaced as to accomplish a takeup of approximately 0.010 inch per tooth.

3.3.5 *Pawl.*—The pawl shall be of good quality steel, properly heat-treated to resist wear or breakage under normal service conditions. The pawl shall be designed to be properly assembled to the tension bands and to lock the ratchet at the desired point.

3.3.6 ***Removable key or wrench.*** — The key or wrench provided to actuate the spool shall be of good quality steel and of sufficient length and strength to actuate the spool in contracting the bands and piston rings to the diameter desired.

3.4 Finish. — The piston ring compressors shall have a smooth finish, free from burrs, roughness and excessive oxide scale.

3.5 Identification marking.—The piston ring compressors shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.6 Size 1, 2⅛- to 5-inch capacity.—Size 1 piston ring compressors shall be capable of servicing pistons ranging from 2⅛ up to and including 5 inches in diameter, and be similar to figure 1.

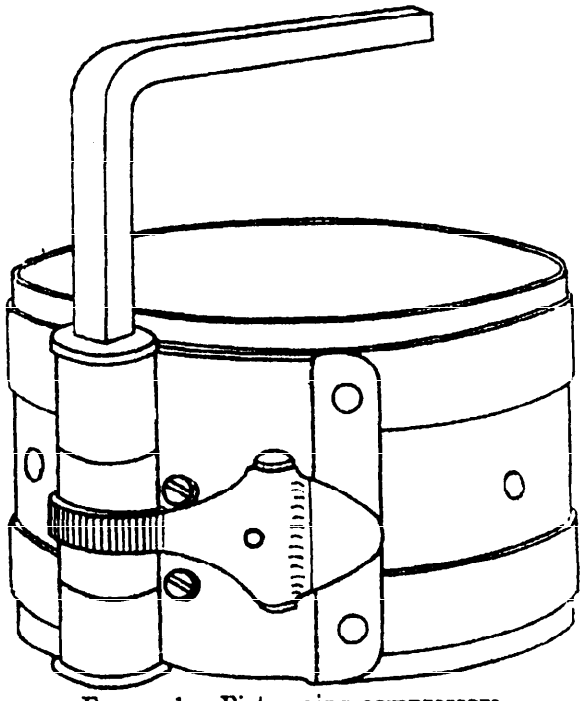


FIGURE 1.—Piston ring compressors.

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3.7 Size 2, 3½- to 7-inch capacity.—Size 2 piston ring compressors shall be capable of servicing pistons ranging from 3½ up to and including 7 inches in diameter, and shall be similar to figure 1.

3.8 Workmanship.—Workmanship shall be of the highest grade throughout and equal in every respect to the best commercial practice. The piston ring compressors shall be free from rust, burrs, fins, and any imperfections which may impair their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All compressors of the same size presented at one time shall be considered a lot for purposes of acceptance, inspection, and tests.

4.1.2 *Sampling for inspection.*

4.1.2.1 *Sampling for intermittent procurement in isolated lots.*—A random sample of compressors shall be taken by the Government inspector in accordance with table I from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Military Standard MIL-STD-105.

TABLE I.—*Sampling for visual and dimensional inspection A.Q.L. (approx.) = 1.5 percent defective*

Number of compressors in inspection lot	Number of compressors in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110 ...	15	0	1
111 to 300 ...	25	1	2
301 to 500 ..	35	1	2
501 to 800 ...	50	2	3
801 to 1300 ..	75	3	4
1301 and over .	110	4	5

NOTE.—The Government inspector may institute tightened or reduced inspection in accordance with Military Standard MIL-STD-105.

4.1.2.2 *Sampling for continuous production.*—Where there is continuous production for the Government and inspection lots include at least 200 items, sampling for Government inspection shall be conducted in accordance with Military Standard MIL-STD-105 at inspection level I and A.Q.L. = 1.5 percent defective; the type of sampling (single, double, or multiple) shall be determined by the Government inspector.

4.1.3 *Sampling for lot acceptance tests.*—A random sample of compressors shall be taken by the Government inspector from each inspection lot in accordance with table II and shall be subjected to the tests specified in 4.3.

TABLE II.—*Sampling for lot acceptance tests*

Number of compressors in inspection lot	Number of compressors in sample	Number of items nonconforming on any lot acceptance tests	
		Acceptance number	Rejection number
15 and under	3	0	1
16 to 40	5	0	1
41 to 110 ...	7	0	1
111 to 300 ...	10	0	1
301 to 800 ...	15	1	2
801 to 1300 ..	25	1	2
1301 and over .	35	2	3

4.2 Visual and dimensional inspection.—Each of the sample compressors taken in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any compressor in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective compressors in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests.—Each of the sample compressors taken in accordance with table II shall be given such tests as may be necessary to verify compliance with this specification. Any sample compressor which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming compressors in any sample exceeds the accept-

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ance number for that sample, the lot represented by the sample shall be rejected.

4.4 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming compressors. The Government inspector shall again examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary requirements.

5.1.1 *Preservation.*—Cleaning and preservation shall be in accordance with the manufacturer's standard practice.

5.1.2 *Packaging.*—Commercial packages will be acceptable.

5.1.3 *Packing.* — The equipment shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose in such manner as to insure acceptance and safe delivery to the designated point by common or other carrier at the lowest rate of the carrier.

5.1.4 *Marking.*

5.1.4.1 *Interior containers.* — Packages and interior containers shall be marked with the name of the material and the name or trademark of the contractor, of such known character that the source of supply may be readily determined.

5.1.4.2 *Exterior containers.* — Shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order, under which shipment is made, the name of the contractor, and the number of the contract or order.

5.2 Military requirements.

5.2.1 *Preparation for shipment.* — Cleaning, preservation, packaging, packing, and marking for shipment shall be in accordance with Military Specification MIL–P–15424.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following and exercise any desired options offered herein:

(a) Title, number, and date of this specification.

(b) Size required (see 1.2).

(c) Whether preparation for domestic shipment—immediate use, domestic shipment and storage, or overseas shipment is required (see section 5).

6.2 Transportation description. — The applicable transportation description is as follows:

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Tools, not otherwise indexed.
Truckload minimum weight 30,000 pounds.

Minimum weights shown apply when material is transported at classification ratings. They will vary when material moves on commodity rates. When in doubt consult General Services Administrational Regional Traffic Manager.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—O E T
Navy—Sh MC Y
Air Force.

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